

# AMMAR HANNUN

AI Engineer | LLM Applications • Machine Learning • Backend Systems

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## PROFILE

Final year Computer Science student at UNSW (AI specialisation) who ships production AI systems end to end: LLM tool calling and RAG, ML pipelines, and Python backends on FastAPI and Django. Two live products built solo, from data pipeline to deployed frontend. Seeking an AI engineering internship or part time role in Sydney, available immediately.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, JavaScript, SQL

**AI & LLM:** OpenAI API, Anthropic Claude API, tool calling, RAG (embeddings, pgvector, TF-IDF), prompt engineering, MCP, agent design

**Machine Learning:** PyTorch, scikit-learn, XGBoost, Random Forest, ResNet50, CNNs, LSTMs, pandas, NumPy, transfer learning, hyperparameter tuning

**Backend & Data:** FastAPI, Django, REST APIs, Celery, Redis, PySpark, Supabase (PostgreSQL, Auth, Row Level Security, pgvector), Docker

**Tooling & Delivery:** Git/GitHub, Vercel, Railway, Docker, pytest, Cursor, Claude Code, OpenAI Codex, Windsurf, v0

## PROJECTS

**Finio** | *AI Personal Finance App* | Python, FastAPI, scikit-learn, Supabase, OpenAI Live • 2026

- Built an end to end app that parses CSV and PDF bank statements across multiple Australian bank formats, then categorises transactions through a hybrid pipeline: deterministic rules first, cached Naive Bayes and TF-IDF for the remainder, with active learning so user corrections become training examples on later uploads.
- Shipped a tool calling AI coach that invokes typed tools (category totals, transaction filters, spend checks, concept lookup) to compute over real transaction data instead of hallucinating figures, grounded by a RAG layer over a curated Australian finance knowledge base using OpenAI embeddings and Supabase pgvector, with TF-IDF fallback so the app stays usable with no API key.
- Added per category z score anomaly detection, statistical recurring bill detection, balance runway forecasting and savings goal projection; secured with Supabase Auth and per user Row Level Security; validated by a 69 case test suite.

**LLM Find & Replace** | *Natural language regex engine at scale* | Django, Celery, Redis, PySpark Live • 2026

- Natural language to regex: a user describes a pattern such as "find email addresses" and an LLM generates the regex, which is validated for compilation and catastrophic backtracking before it is allowed to run.
- Processed a 1,000,000 row CSV in roughly 10 seconds by distributing regex replacement across 8 Spark partitions; Celery and Redis run jobs asynchronously with live progress polling, and a Redis regex cache removes duplicate LLM calls.
- Separated the HTTP layer from the processing layer as distinct Django apps, containerised with Docker Compose and deployed live, with a paged results API so the browser never loads the full dataset.

**ML Benchmarking: xRFM vs XGBoost vs Random Forest** | Python, scikit-learn, XGBoost Apr 2026

- Benchmarked 3 models across 5 tabular datasets (2 classification, 3 regression) with hyperparameter tuning on a held out validation split, evaluating accuracy, AUC, RMSE, training time and per sample inference time.
- Compared 4 feature importance methods (PCA, mutual information, permutation, xRFM AGOP diagonal) and ran learning curve experiments on AUC and training time against training set size to assess scalability.

**Breast Cancer Classification & Segmentation** | PyTorch, ResNet50 Jul 2025

- Fine tuned ResNet50 with ImageNet weights on breast ultrasound images across 3 classes (benign, malignant, normal), reaching 83% test accuracy using augmentation, dropout regularisation, early stopping and learning rate scheduling.
- Built a segmentation model reaching 0.64 IoU to highlight tumour regions, supporting computer aided diagnosis research.

**Also built:** Toohak backend quiz platform (REST API), Java train network simulator and OOP game engine refactor with design patterns and JUnit testing, Huffman compression in C, graph based route planning, and a PyTorch comparison of Linear, MLP, CNN, SRN and LSTM models on sequence prediction.

## EXPERIENCE

**IT / Power Platform Intern** | EDGO Jan – Feb 2026

- Automated internal workplace processes with Power Automate across SharePoint, Outlook and Excel: email notifications, approval chains, spreadsheet updates and trigger based workflows.
- Developed a Power Apps sign up platform backed by SharePoint Lists, and evaluated Microsoft Copilot and AI agent patterns for extending internal workflow tooling.

**Math & Science Tutor** | A2 Tutors Sep 2024 – Present

- Tutor mathematics and science across multiple year levels, building customised study plans and translating complex concepts for non technical audiences.

**Peer Mentor** | UNSW College Sep 2025 – Present

- Support first year students with onboarding, study habits and referrals to support services.

## EDUCATION

**UNSW Sydney** | *Bachelor of Computer Science, Artificial Intelligence Specialisation* 2025 – 2027

Coursework: Machine Learning & Data Mining, Neural Networks & Deep Learning, Computer Vision, AI, Data Structures & Algorithms, Software Engineering, Computer Systems, OOP.

**UNSW College** | *Diploma in Computer Science* 2024

## ACHIEVEMENTS

- **3rd place, Debugged Hackathon** (UNSW Founders x AI Society) for AURA, a campus social feed scored by an LLM judge.
- **Microsoft & LinkedIn Generative AI Certificate** covering generative AI foundations, prompt design and applied LLM workflows.
- **Valedictorian and top of class** at Dar Jana International School, 2023; Student Ambassador, SAT Centre, 2022.